

UDP connection with sockets

Application Note

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1 Applicability Table

Table 1: Applicability Table

Products
ME310G1 SERIES
ME910G1 SERIES
ML865G1 SERIES

2 Introduction

2.1 Scope

The scope of this document is to give information about how to establish a UDP connection to a server and transmit some data by using a Telit module.

In addition, also the possibility to create a UDP socket in listening mode will be presented.

2.2 Audience

This document is intended for system integrators who are using the Telit XX123Z4 module in their products.

2.3 Contact Information, Support

For technical support and general questions, e-mail:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com

Alternatively, use: <https://www.telit.com/contact-us/>

Product information and technical documents are accessible 24/7 on our website:
<https://www.telit.com>

2.4 Conventions

Note: Provide advice and suggestions that may be useful when integrating the module.

Danger: This information MUST be followed, or catastrophic equipment failure or personal injury may occur.

ESD Risk: Notifies the user to take proper grounding precautions before handling the product.

Warning: Alerts the user on important steps about the module integration.

All dates are in ISO 8601 format, that is YYYY-MM-DD.

2.5 Terms and conditions

Refer to <https://www.telit.com/hardware-terms-conditions/>.

2.6 Disclaimer

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3 UDP Quick start

This chapter will present how to establish a UDP connection to a server to send data. To explain a simple UDP connection, the server **modules.telit.com** will be used as an example.

3.1 Initial configuration

The module provides three different ports that allow communication via AT commands: USB, main UART, and aux UART.

Once you connect one of these three ports on your PC, you can use a generic serial communication terminal software to communicate with the module.

In this guide, the Telit Cinterion AT Controller will be used.

Telit Cinterion AT controller

Telit AT controller allows to establish a serial communication and to send AT commands to the Telit Cinterion module.

As shown in Figure 1, it is possible to write AT commands manually or choose from the default sets present in the left panel.

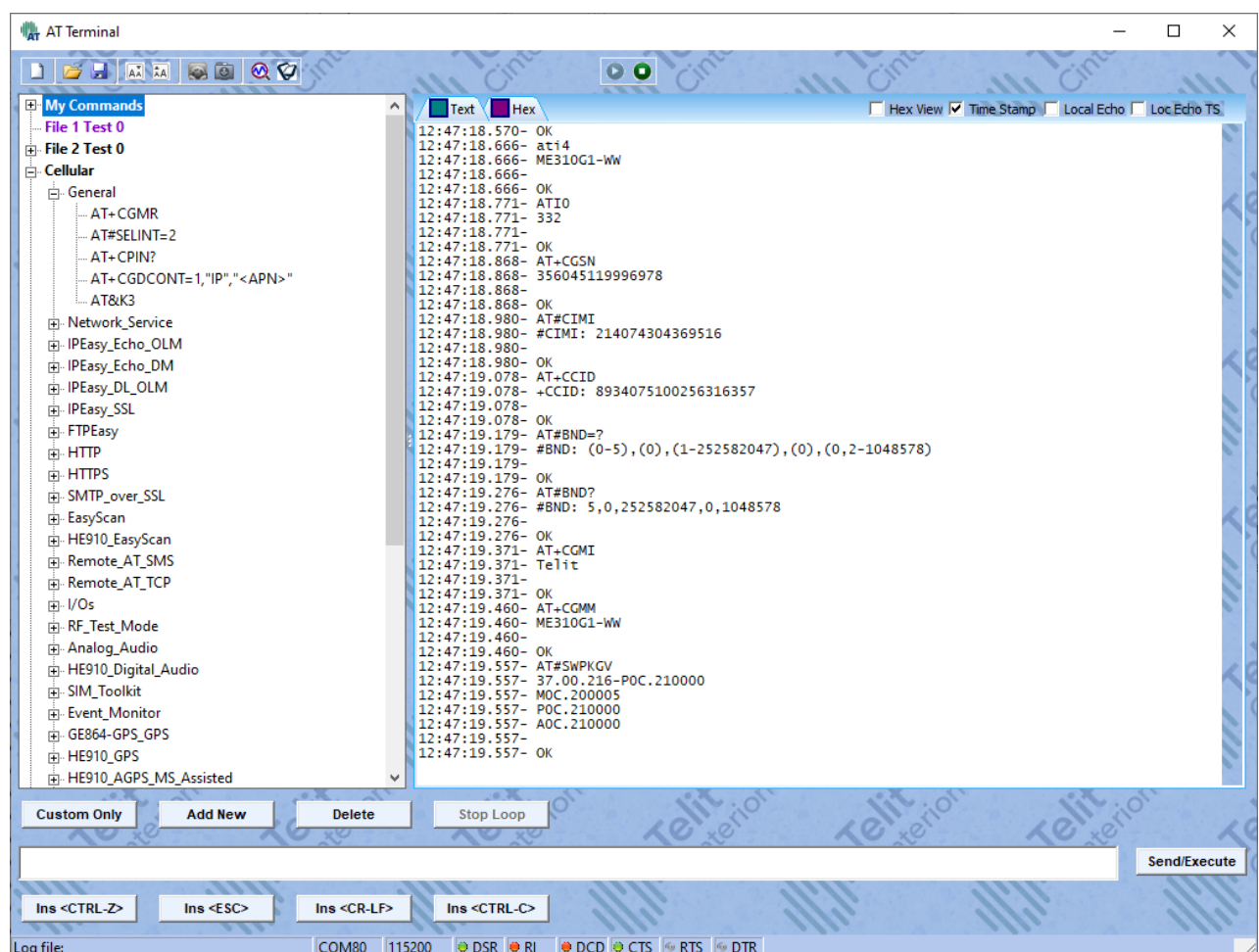


Figure 1 Telit Cinterion AT Terminal

Module registration

Once the module is connected and responsive to AT commands, it is important to verify the registration status with the network. In fact, to establish a UDP connection, the module must be registered (home or roaming) for data traffic services.

The registration status can be verified by querying the **AT+CREG?**, **AT+CGREG?** (PS services valid for 2G or 3G) and **AT+CEREG?** (EPS services valid for LTE).

Commands example:

- Module registered home network

AT+CREG?

+CREG: 0,1

OK

AT+CGREG?

+CGREG: 0,1

OK

- LTE Module registered roaming

AT+CREG?

+CREG: 0,5

OK

AT+CEREG?

+CREG: 0,5

OK

If the module does not report similar results, please check the “80000NT11696A - 2G 3G 4G Registration Process Application Note”.

PDP context activation

If the module is registered to EPS services (LTE) or PS services (2G/3G), it is possible to obtain an IP address with the AT#SGACT command.

You must select the <cid> related to the APN used for data services.

Commands example:

AT+CGDCONT?

+CGDCONT: 1,"IP","nxt17.net","",0,0,0,0

+CGDCONT: 2,"IPV4V6","","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0".0.0,0.0

+CGDCONT: 3,"IPV4V6","", "0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0

+CGDCONT: 4,"IPV4V6","","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0

+CGDCONT: 5,"IPV4V6","","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0

+CGDCONT: 6,"IPV4V6","","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0

OK

The APN has been configured to cid=1, so the context will be requested for this cid.

AT#SGACT=1,1

#SGACT: 10.35.25.68

OK

3.2 Outgoing UDP socket

The server "mobile.telit.com" provides different features based on the port used.

Port	Description
10508	Upload port
10509	Download port
10510	Echo port

In this first example, we will use the echo server to send data and obtain almost instantaneously the same message back.

Since Telit modules allow to open a socket in two different modes (online and command mode) in the following examples each modality is considered individually.

Socket configuration

Before establishing the connection with the server, the socket must be configured. We will use the socket identifier 1 with the default configuration.

AT#SCFG?

#SCFG: 1,1,300,90,600,50

#SCFG: 2,1,300,90,600,50

#SCFG: 3,1,300,90,600,50

#SCFG: 4,1,300,90,600,50

#SCFG: 5,1,300,90,600,50

#SCFG: 6,1,300,90,600,50

#SCFG: 7,1,300,90,600,50

#SCFG: 8,1,300,90,600,50

#SCFG: 9,1,300,90,600,50

#SCFG: 10,1,300,90,600,50

OK

If the configuration for <connId>=1 is not the one represented above and you want to set the default configuration, it is possible to set it with the command below.

AT#SCFG=1,1,300,90,600,50

OK

UDP socket in online mode

Once the socket has been configured, we are ready to establish the connection. This is possible using the socket dial command **#SD**. In this command it is important to set UDP as `<txProt>`, the correct port and hostname of the server, an arbitrary local port, and the `<connMode>` to "online mode connection". The other parameters can be set with their respective default values.

Since online mode has been set, the module will reply with the **CONNECT** URC, and all the data that is written in the AT interface is directly sent to the server. Similarly, all the received data is printed in the AT interface.

```
AT#SD=1,1,10510,"modules.telit.com",0,1000,0
```

CONNECT

```
HelloWorld      ← Data sent
```

```
HelloWorld      ← Data received
```

The "HelloWorld" message sent has been received by the server and sent back to the module.

In online mode, the module does not accept AT commands since they are considered as data to be sent to the server.

For this reason, it is necessary to return the command mode to be able to send again AT commands.

This is possible by using the escape sequence "+++". Once this is sent, the module responds with OK, and the AT commands are effective again.

```
AT#SD=1,1,10510,"modules.telit.com",0,1000,0
```

```
HelloWorld
```

```
HelloWorld
```

```
+++
```

```
OK
```

```
AT
```

```
OK
```

The escape sequence used to return the command mode is sent among the data in a UDP packet. To prevent the escape sequence from being included in the data to be sent, the **#SKIPESC** command must be used.

Note that the escape sequence does not close the connection, but only suspends it. This means that it is possible to return in online mode with the **#SO** command specifying the `<connId>` of the socket enabled. The module will reply with **CONNECT** and the channel is open again.

```
AT#SO=1
```

CONNECT

NewMessage

NewMessage

UDP socket in command mode

The same connection can be opened directly in command mode by setting <connMode> to "command mode connection" on the #SD command.

```
AT#SD=1,1,10510,"modules.telit.com",0,1000,1
```

OK

The module will reply with OK but the connection is opened in the background.

It is possible to send data also in command mode by using #SSLSEND and #SSLSENDEXT commands. The first allows sending an undefined number of data delimited by the Ctrl-Z char (0x1A hex) while the second must require the data length as an input parameter.

```
AT#SSEND=1
```

```
> MyDataString1<CTRL-Z>
```

OK

SRING: 1

```
AT#SSENDEXT=1,13
```

```
> MyDataString2
```

OK

In command mode, the reception of data is notified by the **SRING** URC. The default SRING format notifies the <connId> of the socket on which the data should be read.

It is possible to read the received data by using the **#RECV** command indicating the length of the data to be read.

SRING: 1

```
AT#SRECV=1,13
```

```
#SRECV: 1,13
```

```
MyDataString1
```

OK

Since we are using a UDP connection, it is important to verify the origin of the packet received. In fact, a UDP connection does not require the handshake, so the port is opened to all the possible IP addresses. This is possible for each single datagram enabling the <UDPInfo> flag of #RECV command.

```
AT#SRECV=1,20,1
```

```
#SRECV: "185.86.42.218",10510,1,13,0
```

```
MyDataString1
```

OK

Where the remote IP address, the remote port, the connId, the length of the datagram, and the size of the remaining data to be read are shown.

The same information can be directly visualized in the SRING URC.

This mode can be enabled by acting on the <srMode> of the #SCFGEXT command. This command has effect only if sent before the connection is open.

AT#SCFGEXT=1,3,0,0

OK

AT#SD=1,1,10510,"modules.telit.com",0,1000,1

OK

AT#SSEND=1

> MyDataString1

OK

SRING: "185.86.42.218",10510,1,13,0,MyDataString1

Status of a socket

The status of a socket can be verified with the #SS and #SI commands.

#SS command provides information about the status of the socket, the local and remote IP addresses, and ports.

AT#SS=1

#SS: 1,2,100.104.235.99,1000,185.86.42.218,10510

OK

In this example, socket 1 is in suspended state (<state>=2) since it is in command mode.

On the other hand, the #SI command provides information about the data sent, received, and buffered waiting to be read. The last counter <ack_waiting> value is always 0.

AT#SI=1

#SI: 1,0,0,0,0

OK

AT#SSEND=1

> MyDataString1

OK

SRING: 1

AT#SI=1

#SI: 1,13,0,13,0

←13 bytes have been sent and there are 13 to be read

OK

```
AT#SRECV=1,13
```

```
#SRECV: 1,13
```

```
MyDataString1
```

```
OK
```

```
AT#SI =1
```

```
#SI: 1,13,13,0,0
```

 ← 13 bytes have been sent and 13 have already been read

```
OK
```

3.3 Incoming UDP socket

Telit modules allow to create sockets in listening mode for an incoming UDP connection.

Since the UDP socket in listening mode must be reachable from an external connection, the IP address should be public or the module should be connected to the same private network of the clients.

In the following example, the module will have a public IP.

Open a UDP socket listening

The socket can be created with the #SLUDP command.

```
AT#SLUDP=1,1,1000
```

```
OK
```

From this moment, the module is listening on port 1000 for incoming connections.

We can verify the status of this socket with the #SS command.

```
AT#SS=1
```

```
#SS: 1,4,83.224.144.xx,1000
```

```
OK
```

Where the address and the port enabled are shown.

Accept an incoming connection in online mode

We can simulate an incoming connection using the Netcat application for Windows.

With this, we can establish a UDP connection to our module and send some data.

```
> ncat -u 83.224.144.xx 1000
```

```
> Hello
```

Once this command has been executed, the cmd opens a socket and all the data inserted in the cmd window is directly sent to the connection. For example, we can send the message "Hello".

On the module side, we will receive the SRING: 1 which notifies that an incoming connection has arrived.

We can decide to accept or refuse the connection respectively with the commands AT#SA and AT#SH. Let's accept the connection for the moment.

```
AT#SA=1
CONNECT
Hello
```

Accepting the connection, the module enters online mode ready to send or receive some data. All the data that we sent from the module side is received and printed by the Netcat socket and vice versa.

Accept an incoming connection in command mode

When an incoming connection is accepted with the #SA command, it is opened by default in online mode. However, it is possible to open it also in command mode by specifying the <connMode> flag.

```
SRING: 1
AT#SA=1,1
OK
SRING: 1
AT#SRECV=1,20
#SRECV: 1,6
Hello
OK
```

In this example, after sending the #SA command, the module responds with OK (command mode) and another SRING: 1. The latter notifies that some data is ready to be read. Using the #SRECV command, we can read the incoming message.

Also in this case, we can use the verbose #SRING mode by acting on the <srMode> field of the #SCFGEXT command. In this way, we can see more information about the incoming data arrived.

```
AT#SCFGEXT=1,3,0,0
OK
AT#SLUDP=1,1,1000
OK
SRING: 1
AT#SA=1,1
OK
SRING: "130.41.171.171",37430,1,5,0,TEST
```

Send data in command mode

Also for the incoming UDP socket, it is possible to use #SEND and #SENDEXT commands to send data (in the same way as already explained in 3.2).

However, the UDP connection is connectionless (there is not a handshake between the two parts) and once the UDP port is open, more than one client can be connected to the

module. However, both #SSEND and #SSENDEXT allow to send of the data to the last client to which the module has received some data.

To deal with this problem, #SSENDUDP and #SSENDUDPEXT commands allow to send data specifying the remote IP address and port.

We can try opening two different remote UDP clients using two Netcat sessions. This will result into two different connections, with the same IP addresses but different ports.

```
AT#SLUDP=1,1,1000
```

```
OK
```

```
SRING: 1
```

```
AT#SA=1,1
```

```
OK
```

```
SRING: "130.41.171.171",15087,1,7,0,CLIENT1
```

← Data from CLIENT1

```
SRING: "130.41.171.171",16783,1,7,0,CLIENT2
```

← Data from CLIENT2

```
AT#SSENDUDP=1,130.41.171.171,15087
```

← Data to CLIENT1

```
> Client1_send
```

```
OK
```

```
AT#SSENDUDPEXT=1,12,130.41.171.171,16783
```

← Data to CLIENT2

```
> Client2_message
```

```
OK
```

Firewall

Telit modules provide an internal firewall that controls the access of the incoming connections. All packets not matching the rules are silently discarded. The firewall applies to incoming (listening) connections. different remote.

The #FRWL command allows to add or remove an accept chain to/from the internal firewall.

Let's suppose that we want to establish a connection only from a specific IP address. We can add this IP address to the accept chain. All the packets received from other IP addresses are discarded.

```
AT#FRWL=1, "130.41.171.171","255.255.255.255"
```

It is important to send this command before opening the UDP socket in listening.

4 Acronyms and Abbreviations

Table 2: Acronyms and Abbreviations

Acronym	Definition
ADC	Analog – Digital Converter
CLK	Clock
CMOS	Complementary Metal – Oxide Semiconductor
CS	Chip Select
DAC	Digital – Analog Converter
DTE	Data Terminal Equipment
ESR	Equivalent Series Resistance
GPIO	General Purpose Input Output
HS	High Speed
HSDPA	High-Speed Downlink Packet Access
HSIC	High-Speed Inter Chip
HSUPA	High-Speed Uplink Packet Access
I/O	Input Output
MISO	Master Input – Slave Output
MOSI	Master Output – Slave Input
MRDY	Master Ready
PCB	Printed Circuit Board
RTC	Real-Time Clock
SIM	Subscriber Identification Module
SPI	Serial Peripheral Interface
SRDY	Slave Ready
TTSC	Telit Technical Support Centre
UART	Universal Asynchronous Receiver Transmitter
UMTS	Universal Mobile Telecommunication System
USB	Universal Serial Bus
VNA	Vector Network Analyzer
VSWR	Voltage Standing Wave Ratio
WCDMA	Wideband Code Division Multiple Access

5 Related Documents

Refer to <https://dz.telit.com/> for current documentation and downloads.

Table 3: Telit Related Documents

S.no	Book Code	Document Title
1	80617ST10991A	ME310G1/ME910G1/ML865G1 AT Commands Reference Guide
2	80000ST10028A	IP Easy User Guide
3	80000NT11696A	2G 3G 4G Registration Process

6 Document History

Table 4: Document History

Revision	Date	Changes
0	2024-03-12	Initial release

From Mod.0809 Rev.9

